GAST-PF: missing-aware spatiotemporal encoding with decoupled-query future decoding

N agents T = 8 history frames T_f = 12 future steps K = 20 modes C = 128 channels ⊙ multiply ⊕ add

Observed trajectories

Complete trajectory
focal + neighbours

T
N

Binary validity mask m [B, N, T]

Feature construction
x = [Δp, Δv, m] [B, N, T, 4]

Target convention
actor slot 0 = focal; neighbours = 1:N

GAST-PF encoder

forward / reversed missing-run deltas δ

Validity mask m
[B, N, T]

Missing-run deltas
δᶠ and δᵇ from m

Temporal Decay
γ = exp(−ReLU(Wδ))

decay gate γ

History features
8 × 0.4 s [B, N, T, 4]

Agent tokens

Patch embedding
Linear → LayerNorm → LeakyReLU

GSM spatial attention
ghost-token pooling + agent MHA

Spatial embeddings s
[B, T, N, C]

⊙

Bidirectional Temporal Mamba

4 blocks per direction; forward + reversed, summed

Last-step readout
take y[:, −1]

Actor embeddings h [B, N, C]

Default: gast_anchor_geo = false; padding actors are masked throughout.

Structured future decoder with decoupled queries

Neighbour prediction head
x_encoder[:, 1:N] → ŷ_others

Actor embeddings h
[B, N, C]

Mode query Q_m
h_0 + learnable mode embeddings

Future time embedding
MLP(t), t = 0.4 ... 4.8 s

State query Q_s
[B, T_f, C]

Mode localization module

scene context h

Cross-Attn → **Self-Attn** → **Mode head**

ŷ [B, K, T_f, 2] + π + scale

State consistency module

scene context h

Cross-Attn → **Bi-Mamba** → **State head**

dense prediction [B, T_f, 2]

$Q_h = Q_m \oplus Q_s$

Hybrid coupling

Cross-Attn
Self-Attn
Mode Self-Attn
Bidirectional Mamba
Coupled trajectory head

mode × state interaction

ŷ_new [B, K, T_f, 2]
+ π_new + scale_new

Training objectives: neighbour regression + dense regression + multimodal regression / classification + Laplace NLL